

Outlier Identification and Management

Finding Outliers in Excel

Several methods, one goal: identify unusual values that may need investigation

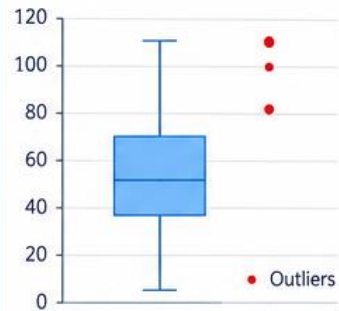
1

Visual inspection

Box plot or scatter plot

Outliers often appear as isolated points in a box plot or scatter plot.

Box plot (example)



In Excel

1. Select the data.
2. Go to Insert → Statistical Chart → Box & Whisker.
3. Look for points outside the whiskers (these are potential outliers).



Available in Excel 2016 and later (Insert → Insert Statistic Chart).

2

Z-score method

Values far from the mean

An observation is an outlier if its z-score is typically greater than 3 (in absolute value).

$$z = \frac{x - \text{mean}}{\text{standard deviation}}$$

Value	Z-score	Outlier?
52	0.12	No
55	0.35	No
58	0.62	No
120	3.48	Yes
49	-0.21	No

In Excel

1. Calculate the mean (AVERAGE) and standard deviation (STDEV.S).
2. Compute the z-score for each value.
3. Flag as outlier if $\text{ABS}(z) > 3$.

Excel formula (in cell C2)

```
= (A2 - AVERAGE($A$2:$A$100)) /  
STDEV.S($A$2:$A$100)
```

3

IQR method

Values outside a typical range

Outliers are values below $Q1 - 1.5 \cdot \text{IQR}$ or above $Q3 + 1.5 \cdot \text{IQR}$.

$$\text{IQR} = Q3 - Q1$$

$$\text{Lower} = Q1 - 1.5 \cdot \text{IQR}$$

$$\text{Upper} = Q3 + 1.5 \cdot \text{IQR}$$

Value	Outlier?
18	No
21	No
22	No
23	No
85	Yes
90	Yes

In Excel

1. Compute Q1 and Q3 using QUARTILE.
2. Calculate IQR, lower and upper bounds.
3. Flag values outside the bounds.

Excel formulas (example range A2:A100)

```
Q1: =QUARTILE.INC(A2:A100,1)  
Q3: =QUARTILE.INC(A2:A100,3)  
IQR: =Q3 - Q1  
Lower: =Q1 - 1.5*IQR  
Upper: =Q3 + 1.5*IQR  
Flag (in B2): =IF(OR(A2<Lower,A2>Upper),  
"Yes","No")
```

4

Modified Z-score

More robust (uses median)

Uses the median and the Median Absolute Deviation (MAD). Consider an outlier if $|M| > 3.5$.

$$M = 0.6745 \cdot \frac{x - \text{median}}{\text{MAD}}$$

Value	Modified Z	Outlier?
18	-0.12	No
21	0.05	No
22	0.18	No
23	0.31	No
85	4.21	Yes
90	5.03	Yes

In Excel

1. Compute the median (MEDIAN).
2. Calculate the MAD (median of absolute deviations).
3. Compute the modified z-score.
4. Flag as outlier if $\text{ABS}(M) > 3.5$.

Excel formula (in B2)

```
=0.6745*(A2-MEDIAN($A$2:$A$100))/  
MEDIAN(ABS($A$2:$A$100)-MEDIAN($A$2:$A$100))
```

5

Domain knowledge

Business rules

Sometimes you know what values are unrealistic based on context (e.g. negative age, very high income).

Value	Rule	Outlier?
-5	Age < 0	Yes
220	Age > 120	Yes
35	—	No
105	—	No

In Excel

1. Define business rules (e.g. min/max allowed values).
2. Use IF statements to flag violations.

Excel formula (example for Age in A2)

```
=IF(OR(A2<0,A2>120),"Yes","No")
```



Tip

Combine multiple methods: use statistical techniques to detect unusual values and domain knowledge to validate them.

Next steps

Identify outliers



Investigate
(e.g. data entry error?)



Decide what to do
(remove, correct, or keep)



Document your decision

Key takeaways

- ✓ There is no single best method.
- ✓ Visualize your data.
- ✓ Use statistical methods and domain knowledge.
- ✓ Always investigate before removing outliers.
- ✓ Document your choices.

Synthetic CSV Dataset with Outliers

1	CustomerID, Age, Income_USD, Monthly_Spend_USD, Debt_USD, City
2	C001, 34, 42000, 1800, 6500, Turin
3	C002, 41, 51000, 2200, 7200, Milan
4	C003, 29, 39000, 1600, 4800, Rome
5	C004, 37, 47000, 2100, 5900, Naples
6	C005, 45, 56000, 2400, 8100, Genoa
7	C006, 32, 44000, 1750, 5300, Bologna
8	C007, 38, 49500, 2050, 6100, Florence
9	C008, 27, 36000, 1500, 4200, Palermo
10	C009, 143, 52000, 2300, 7000, Venice
11	C010, 36, 450000, 2600, 8500, Turin
12	C011, 33, 46000, 12000, 6400, Milan
13	C012, 40, 54000, 2150, -2500, Rome
14	C013, 31, 43000, 1850, 5600, Naples
15	C014, 39, 50500, 2250, 6900, Genoa

Outlier:
Unrealistic age (143)

Outlier:
Extremely high income

Outlier:
Very high monthly spend

Outlier:
Negative debt value

bank_customers_outliers.xlsx (participants): 1,000 customers × 22 columns

(demographics, income, balances, transactions, card, mortgage, score, complaints) with filters and a frozen header, plus a **Data Dictionary** sheet with the plausible range for each column.

bank_customers_outliers_TRAINER.xlsx: same data plus three sheets:

- Statistics** — mean, median, standard deviation, quartiles, IQR fences and z-score thresholds for the 15 numeric columns, all live formulas on the Customers sheet
- Checks** — 13 row-level checks (plausible age, tenure vs age, card spend vs limit, mortgage vs payment, dates, duplicates, score out of range...) with a summary of the counts
- Outlier_Key_Trainer** — the 95 injected anomalies, one per row, with ID, Excel row, columns involved, why it is "to be verified" and how to spot it in Excel

The 8 categories: **A** plausible extreme values (an executive earning €1.85M, a 97-year-old), **B** impossible values (age -3 and 187, a €32M income with extra zeros, an income of "42" in thousands), **C** sentinels and blanks (-999, 9999, empty cells), **D** cross-column inconsistencies (28 years of tenure at age 21, mortgage payment above monthly income, a student earning €240k), **E** future dates and "zero" dates (1900-01-01, 1970-01-01), **F** categorical variants ("lombardia", "Lombardia " with a trailing space, "Male", "Counter"), **G** exact duplicates and near-duplicates with a different ID, **H** a cluster of 12 accounts opened in the same month with 48 transactions of €199 each — normal one by one, anomalous as a group.

A point that lands well in class: on income, the z-score flags 1 outlier only (the €32M inflates the standard deviation and masks everything else), while the IQR rule finds 74. That's the perfect argument for removing sentinels and gross errors first, and only then looking for the "real" outliers.

Dates are formatted as YYYY-MM-DD to stay unambiguous for an international audience; if the participants use US Excel and you'd rather have MM/DD/YYYY, I can switch it.



Watch out for the separator: ; versus , (and comma versus point as the decimal separator) depends on the Windows regional format, not on Office, so if you want the screen to show exactly =IF(A2>1.5, ...) you also need to change the regional format in Settings > Time & Language > Region.

In Windows 11, the path is:

Start → Settings → Time & language → Language & region

Here you can find **Regional format**. The selected format, for example **Italian (Italy)** or **English (United States)**, affects Excel, especially:

-  or  as the decimal separator
-  or  as the formula argument separator

For example:

Italian (Italy):

```
</> excel
```

```
=IF(A2>1,5; "Yes"; "No")
```

English (United States):

```
</> excel
```

```
=IF(A2>1.5, "Yes", "No")
```

If you want to change the **separators specifically**, the most direct path in Windows is:

Control Panel → Clock and Region → Region → Additional settings

There you can change the **Decimal symbol** and the **List separator**.

In Windows 10,

Start → Settings → Time & Language → Region

Then look for **Regional format**.

If you want to change the separators specifically:

Control Panel → Clock and Region → Region → Additional settings

There you can set:

- **Decimal symbol**
- **List separator**

So the wording **"Settings → Time & Language → Region"** is correct for Windows 10.

Missing Values Identification and Management

What are missing values?

Real data is rarely complete. Missing values are common, and it's important to recognise and handle them.

Look at this small dataset:

Customer_ID	Age	Region	Balance	Occupation
C001	25	North	1,200	Engineer
C002		South	850	Teacher
C003	37	NA	2,300	
C004	29	East	N/A	Doctor
C005	NULL	West	670	Artist
C006	45		1,100	Manager
C007	52	Central	?	Retired

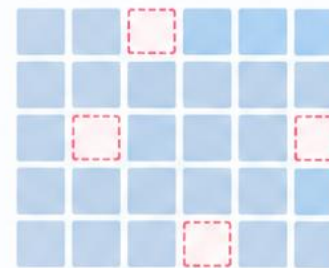
Do you notice anything unusual?



Missing values are data entries that are absent, unknown, not recorded, or not available. They are often represented as blank cells or markers such as NA, N/A, NULL, or ?.

Missing values are pieces of information that should be there, but are not.

Missing values create gaps in our data...



Gaps mean missing information.

They can also appear in real-world forms, such as incomplete records...

Customer Profile
Name:
Age:
Region:
Occupation:
Balance:

Some information is missing here too.

Why they matter



can bias summaries



can distort charts and correlations



can reduce usable sample size



must be handled explicitly

Three types of missing values

How they differ, what they mean, and how to handle them

MCAR

Missing Completely At Random



Definition

The probability that a value is missing is unrelated to the variable itself and unrelated to other observed variables.



Intuition / example

Example: a few survey answers are missing because of a random technical glitch.

ID	Age	Income	Score
1	34	—	88
2	—	52	73
3	28	61	—
4	—	—	90
5	41	48	76

Missing values are scattered randomly across the data.



How to handle

Often the least problematic case. You can use listwise deletion when the amount is small, or simple imputation when appropriate.

Lowest risk of bias



MCAR:

random missingness

MAR

Missing At Random



Definition

The probability that a value is missing depends on other observed variables, but not on the missing value itself once those variables are considered.



Intuition / example

Example: income is more often missing for younger customers; age is observed.

ID	Age	Income	Score
1	22	—	78
2	25	—	85
3	34	56	90
4	47	62	88
5	52	71	76

Missing values in income are more common for younger customers (age is observed).



How to handle

Model-based methods and imputation are usually appropriate: mean within groups, regression imputation, or multiple imputation.

Manageable if related observed variables are available



MAR:

depends on observed data

MNAR

Missing Not At Random



Definition

The probability that a value is missing depends on the unobserved value itself, or on unobserved factors.



Intuition / example

Example: customers with very high debt are less likely to report it.

ID	Debt	Income	Score
1	5	42	80
2	12	58	76
3	28	61	82
4	—	—	74
5	—	—	68

Missing values are concentrated among extreme (typically high) values of the variable itself (e.g., high debt).



How to handle

Most difficult case. Investigate the data-generation process, use domain knowledge, sensitivity analysis, and specialized models when possible.

Highest risk of bias



MNAR:

depends on missing/unobserved data



Key takeaway: the right treatment depends on **why** data is missing, not just on how much is missing.

3 Ways to Handle Missing Values (MV) in Excel

Choose the right method based on your data, the amount of missing values and your analysis goals.

1 Delete rows with missing values (Listwise deletion)

Remove every row that contains at least one missing value.

When to use

- Few missing values
- Missing values are completely random (MCAR)
- You want a simple and clean dataset

	A	B	C	D
1	ID	Age	Income	City
2	1	25	50000	Rome
3	2		62000	Milan
4	3	34		Turin
5	4	28	58000	Naples

How to do it in Excel

1. Select the data range.
2. Go to Data → Filter.
3. In each column, filter "(Blanks)" and delete the corresponding rows.



In a dynamic way (newer Excel versions), you can use the **FILTER** function:

```
=FILTER(A2:D100, (A2:A100<>"")*(B2:B100<>"")*(C2:C100<>"")*(D2:D100<>""))
```

2 Impute with the column mean or median

Replace missing values with a simple statistic of the column (mean or median).

When to use

- Small to moderate amount of missing values
- Numeric variables
- You need a quick and simple solution

	A	B	C
	ID	Age	Income
1	1	25	50000
2	2		62000
3	3	34	
4	4	28	58000

How to do it in Excel

Use IF to replace blanks with the mean or median of the column.

Using the mean:

```
=IF(B2="", AVERAGE(B:B), B2)
```

If B2 is empty, replace it with the column mean.

Using the median:

```
=IF(B2="", MEDIAN(B:B), B2)
```

If B2 is empty, replace it with the column median.



Create a new column with the imputed values to keep the original data.

3 Use a predictive model to estimate missing values

Predict missing values using other variables in the dataset (e.g. regression, machine learning).

When to use

- Moderate to large amount of missing values
- Missing values depend on other variables (MAR or MNAR)
- You want more accurate estimates

	A	B	C	D
1	ID	Age	Income	Experience
2	1	25	50000	2
3	2		62000	5
4	3	34	54000	8
5	4	28	58000	4

How to do it in Excel (example: linear regression)

1. Go to Data → Data Analysis → Regression.
2. Use the other variables (e.g. Income, Experience) to predict the variable with missing values (e.g. Age).
3. Excel will output the coefficients (intercept and slopes).
4. Use the coefficients in a formula to estimate missing values.

Example formula (after running a regression for Age):

```
=IF(B2="", $E$1+$E$2*C2+$E$3*D2, B2)
```

If B2 is empty, predict Age using Income (C2) and Experience (D2).



This approach usually gives better results than simple imputation, but it requires more work and domain knowledge.

There is no single best method.

Consider the amount of missing values, the reason why they are missing, and the goal of your analysis.

The two files are ready (same 22 predictors as the outlier dataset, new seed, everything in English):

bank_customers_missing.xlsx (participants) — **Customers**: 1,000 customers with 1,598 effectively missing cells (7.3%) under every disguise: empty cells, cells of only spaces, text placeholders (N/A, NA, n.d., ?, -, Unknown), numeric sentinels (-999, 9999, 0, date 1970-01-01), numbers saved as text, 4 ghost records (only the ID), 6 almost-empty records, and structural blanks (no card, no mortgage, dormant account, "NULL = 0 complaints"). Three mechanisms: MCAR, MAR (income missing at 28% for self-employed and business owners vs 3% for the others) and MNAR (the savings balance is missing all the more the higher it is). **Data_Dictionary** documents only the "official" conventions (0 = no card, etc.): everything else is to be discovered.

bank_customers_missing_TRAINER.xlsx — same data plus the workflow sheets, all with live formulas:

- **Codes**: the single list of placeholders, read by all the formulas (add a code and the file updates)
- **Missing_Profile**: per column COUNTBLANK, spaces, placeholders, sentinels, text-numbers, % missing, structural share, what to do + chart
- **Missing_Matrix**: 0/1 map with conditional formatting, count per row, ghost / mostly empty flags
- **Clean**: a single formula pattern per column that normalizes everything to a true blank and converts the text-numbers with VALUE
- **Imputed**: the strategies side by side, always next to the clean value and the missing flag (median, mean, group median with VLOOKUP, derivation with DATEDIF, mode, "Unknown" category, spend/limit ratio, constant 0, listwise on 21 vs 8 columns)
- **Strategies**: reference values, missing rate by occupation, and the trainer section that measures each strategy against **Ground_Truth**
- **Missing_Key_Trainer**: key to everything that was injected, cheat sheet of the strategies, classroom outline with the live numbers

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Punti che funzionano in aula (numeri già calcolati nel file): la listwise deletion tiene il 29,5% delle righe (72,3% sulle sole 8 colonne del modello); sul reddito i casi completi danno media 34.752 contro 35.827 vera, la media imputata schiaccia la deviazione standard, la mediana per occupazione recupera 35.514 con metà dell'errore; sul risparmio MNAR le celle mancanti valgono in media 43.669 e noi imputiamo 10.653; la tenure derivata dalla data è esatta 22 volte su 22; la moda del canale indovina il 31%; "NULL = 0 reclami" sbaglia 13 volte su 214.

Allego anche lo script generatore (seed fisso, riproduce entrambi i file). Una nota: nella chat «Dataset outlier per analisi bancaria in Excel» risultava già generata una coppia con questo stesso nome; questa è rifatta da zero, quindi usa i due file insieme — la chiave del trainer vale solo per questo `Customers`.